

Bayesian methods for uncertainty quantification

Part I – Bayesian inference

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"So, you said "*bayesian*"?"

*"Everyone uses Bayesian inference when it is clearly appropriate.
A Bayesian is someone who uses Bayesian inference
even when it might seem inappropriate."*

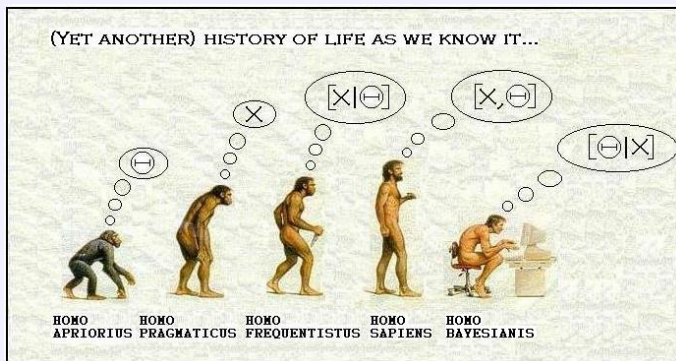
A. Gelman & C. P. Robert,
"Not Only Defended But Also Applied":
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“Bayesianism” is not

- a cult, an attitude/posture
- (only) a controversy (frequentists vs. Bayesians)

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- (only) “a framework”
- (only) the application of Bayes' theorem
(rather set up by Laplace)
- (only) a spam filter
- (only) a network (Bayes net), an algorithm (MCMC, VB, VB-EM)
- recent, even though the rise of Bayesian methods coincides with the advent of computers, often necessary for Bayesian computation.

Bayesian inference

“A simple mathematical theory that characterizes plausible reasoning in the presence of uncertainty.” [(translated) S. Dehaene, Collège de France, 2012]

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Outline

Estimation: generalities and “frequentist” approach

- Point estimation

- Maximum likelihood estimation

Bayesian estimation

- Toy example

- Bayesian paradigm

- Derivation of Bayesian estimators

- Key quantities

- Toy example (cont'd)

- Uncertainty quantification

- Hierarchical Bayesian models

Application to Bayesian inversion/regression

- Probabilistic formulation

- Bayesian regularization

- Generalized Tikhonov regularization

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Problem statement

Let Y be a random variable whose distribution¹ $f(y|\theta)$ depends on an unknown parameter θ .

Definitions

An **n -sample** of Y is an n -tuple $(Y_1, Y_2, \dots, Y_n)$ such that Y_i have the same distribution as Y and are independent.

A **realization** of the sample is an n -tuple $(y_1, y_2, \dots, y_n)$ of values taken by the sample.

Problem

Using a n -sample drawn from $f(y|\theta)$, we aim to best estimate the true value of the parameter θ .

Important remark

Let a problem related to a set $(y_1, y_2, \dots, y_n)$ of n observations (or measurements, data...) whose values are assumed to be driven by an unknown quantity θ (here in the *parameter*) to be *inferred*. Statistical inference about θ requires to postulate a **probabilistic model** $f(\cdot|\theta)$ underlying this set.

¹In this presentation, for the sake of conciseness, $f(\cdot|\cdot)$ will denote a probability density function characterizing a real-valued random variable. The results easily extend to a discrete random variable defined by probabilities.

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Estimator, estimate and estimation

Definition

An **estimator** $\hat{\theta}$ is a random variable $\hat{\theta}(Y_1, \dots, Y_n) = \hat{\theta}_n$, which is a (sufficiently) regular function of the random variables $Y_1, \dots, Y_n$.

Definition

The (numerical) value of $\hat{\theta}$ obtained after replacing the Y_i 's by the realizations y_i (observed values, measurements, data) of Y_i is an **estimate** of θ , i.e., it is expected to provide an approximate value of the unknown parameter θ .

One has conducted an **estimation** of θ .

Example

Let $Y_1, \dots, Y_n$ a sample such that $Y_i \sim \mathcal{N}(m, \sigma^2)$. Then

$$\hat{m} = \frac{1}{n} \sum_{i=1}^n Y_i$$

is an estimator of m .

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Quality of an estimator

Two main figures-of-merit

- Bias free

$$\mathbb{E} \left[\hat{\theta}_n \right] = \theta$$

↪ no systematic error

- Convergence, which can be translated as

$$\lim_{n \rightarrow +\infty} \text{var} \left[\hat{\theta}_n \right] = 0$$

↪ more observations, less uncertainty

Mean square error

$$\begin{aligned} e_n^2(\theta) &= \mathbb{E} \left[\left(\hat{\theta}_n - \theta \right)^2 \right] \\ &= \text{var} \left[\hat{\theta}_n \right] + \left(\mathbb{E} \left[\hat{\theta}_n \right] - \theta \right)^2 \end{aligned}$$

Quality of an estimator

Example

Let $Y_1, \dots, Y_n$ a sample such that $Y_i \sim \mathcal{N}(m, \sigma^2)$. Let define two estimators of m

$$\hat{m}_1 = Y_1 \quad \text{and} \quad \hat{m}_2 = \frac{1}{n} \sum_{i=1}^n Y_i.$$

Then

$$\mathbb{E}[\hat{m}_1] = m \quad \text{var}[\hat{m}_1] = \sigma^2$$

$$\mathbb{E}[\hat{m}_2] = m \quad \text{var}[\hat{m}_2] = \frac{\sigma^2}{n}$$

→ $\hat{m}_2$ “better” than $\hat{m}_1$.

Cramer-Rao bound

Property

Let $\hat{\theta}$ an unbiased (bias-free) estimator of θ . Then one defines the lower bound

$$\text{var} \left[\hat{\theta} \right] \geq \text{BCR}_n(\theta) \triangleq \mathcal{I}(\theta)^{-1}$$

where $\mathcal{I}(\theta)$ is the **Fisher information**

$$\mathcal{I}(\theta) = \text{E} \left[- \frac{\partial^2 \log f(Y_1, \dots, Y_n | \theta)}{\partial \theta^2} \right].$$

Definition

An unbiased estimator is **efficient** if $\text{var} \left[\hat{\theta} \right] = \text{BCR}_n(\theta)$.

Property

An efficient estimator is unique.

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Maximum likelihood estimator

Definition

Let $Y_1, \dots, Y_n$ a n -sample such that $Y_i \sim f(y|\theta)$. The **likelihood** (function) is the function $\mathcal{L}_n$ defined by

$$\mathcal{L}_n(y_1, \dots, y_n; \theta) = f(y_1, \dots, y_n | \theta).$$

Remark

Let $(y_1, y_2, \dots, y_n)$ a realization. Then $\mathcal{L}_n(y_1, \dots, y_n; \theta)$ **measures the fit** between the observations/measurements/data and the probabilistic model $f(\cdot|\theta)$ (assumed beforehand to describe them).

Definition

The maximum likelihood (ML) estimator $\hat{\theta}_{\text{ML}}$ of the parameter θ is

$$\hat{\theta}_{\text{ML}} = \underset{\theta}{\operatorname{argmax}} \mathcal{L}_n(Y_1, \dots, Y_n; \theta)$$

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Maximum likelihood estimator

Property

- Asymptotically unbiased

$$\lim_{n \rightarrow +\infty} \mathbb{E} [\hat{\theta}_n] = \theta$$

- Convergent

$$\lim_{n \rightarrow +\infty} \text{var} [\hat{\theta}_n] = 0$$

- Asymptotically efficient

$$\lim_{n \rightarrow +\infty} \frac{\text{var} [\hat{\theta}_n]}{\text{BCR}_n(\theta)} = 1$$

- Asymptotically normal

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta)^{-1})$$

Maximum likelihood estimator

In practice

In general, the search for the maximum likelihood estimator $\hat{\theta}_{\text{ML}}$ is done by minimizing the neg-log-likelihood

$$\operatorname{argmax}_{\theta} \mathcal{L}_n(Y_1, \dots, Y_n; \theta) = \operatorname{argmin}_{\theta} (-\log f(Y_1, \dots, Y_n | \theta))$$

Example

Let $Y_1, \dots, Y_n$ a n -sample such that $Y_i \sim \mathcal{N}(m, \sigma^2)$. Then

$$\mathcal{L}_n(Y_1, \dots, Y_n; m) = \left(\frac{1}{2\pi\sigma^2} \right)^{\frac{n}{2}} \exp \left(-\frac{\sum_{i=1}^n (Y_i - m)^2}{2\sigma^2} \right).$$

Maximizing the likelihood can be rewritten as

$$\operatorname{argmin}_m \sum_{i=1}^n (Y_i - m)^2$$

which leads to

$$\hat{m}_{\text{ML}} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

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A motivating toy, yet real-world, far-fetched example

Context: quality control

- Let consider a manufacturing factory producing microprocessors (μp).
- The production line is expected to deliver μp with clock frequency $\theta_0 = 20\text{Ghz}$.
- One conducts **measurements** from a set of n distinct μp produced by the line.
- This gives a set of real values $y_1, \dots, y_n$.

One wants to monitor the quality of the production line, i.e., by assessing the clock frequency of the produced μp .

Probabilistic modeling

- The measurements are modeled as realizations of random variables $Y_1, \dots, Y_n$.
- The random variables Y_i are given a probabilistic model parametrized by the unknown μp clock frequency θ , considered as an **unknown parameter**.
- Formally, $Y_i \sim f(y|\theta)$ and, for instance,

$$Y_i|\theta \sim \mathcal{N}(\theta, \sigma^2)$$

where σ^2 models the uncertainty related to the measurement protocol (i.e., noise).

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Statistical estimation

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- Maximum likelihood estimation gives

$$\hat{\theta}_{\text{ML}} = \frac{1}{n} \sum_{i=1}^n Y_i$$

Limits

- This modeling does not reflect the intrinsic variability (e.g., over time) of the production line (i.e., each individual processor is assumed to have exactly the same clock frequency θ)
- This modeling does not capitalize on the knowledge available before any measure (i.e., $\theta \approx \theta_0$).

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Alternative probabilistic modeling

- Uncertainty due to the measurement protocol

$$Y_i|\theta \sim \mathcal{N}(\theta, \sigma^2)$$

- Uncertainty due to the intrinsic variability

$$\theta|\theta_0 \sim \mathcal{N}(\theta_0, \sigma_0^2)$$

where

- the mean θ_0 capitalizes on the knowledge available before any measurement
- the variance σ_0^2 models the uncertainty related to the intrinsic variability of the production line

Paradigmatic shift

From now, the parameter θ is also considered as a random variable!

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Bayesian paradigm

A philosophy...

Principle

Bayesian estimation involves considering the unknown quantity as the realization of a **random variable** θ . This random variable is described by a probabilistic model $f(\theta)$, referred to as the **prior distribution**, which summarizes the information about the parameter θ available before any experiment, i.e., before collecting any observations/data/measurement.

A prior distribution?

- a way to exploit available information for the target problem
 - a probabilistic model of uncertain knowledge
 - a personal or subjective description of this knowledge
 - a necessity when non-Bayesian estimation fails (see later)
 - correction/regularization of the information provided by observations
 - a probabilistic interpretation of regularization/penalization to avoid overfitting
- ... but (okay), also an *pretext* to enter into the "**Bayesian framework**".

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Bayesian paradigm

A philosophy... but not a belief!

"Once again, for Bayesians as much as for any other statistician, parameters are (typically) fixed but unknown. It is the knowledge about these unknowns that Bayesians model as random."

A. Gelman & C. P. Robert,
"Not Only Defended But Also Applied":
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Derivation of Bayesian estimators

Principle

A Bayesian estimator $\hat{\theta}_{\text{Bayes}}$ minimizes a so-called Bayesian risk

$$\hat{\theta}_{\text{Bayes}} = \min_{\hat{\theta}} E \left[c \left(\theta, \hat{\theta} \right) \right]$$

- $c(\theta, \cdot)$ is a cost function which quantifies the error made when estimating θ with $\hat{\theta}$
→ gives access to a possibly wide range of estimators (designed w.r.t. to an appropriate application)
- $E[\cdot]$ should be understood as the expectation over the random variables $Y_1, \dots, Y_n$ **and** the random parameter θ

$$\begin{aligned} E \left[c \left(\theta, \hat{\theta} \right) \right] &= E_{Y_1, \dots, Y_n, \theta} \left[c \left(\theta, \hat{\theta} \right) \right] \\ &= E_{\theta} \left[E_{Y_1, \dots, Y_n} \left[c \left(\theta, \hat{\theta} \right) \mid \theta \right] \right] \\ &= E_{Y_1, \dots, Y_n} \left[E_{\theta} \left[c \left(\theta, \hat{\theta} \right) \mid Y_1, \dots, Y_n \right] \right] \end{aligned}$$

Classical Bayesian estimators

Minimum mean square error

One chooses the quadratic cost

$$c(\theta, \hat{\theta}) = (\theta - \hat{\theta})^2.$$

The risk is the mean square error

$$\text{MSE} = \text{E} \left[(\theta - \hat{\theta})^2 \right].$$

One shows that the estimator minimizing the MSE (MMSE) is

$$\hat{\theta}_{\text{MMSE}} = \text{E} [\theta | Y_1, \dots, Y_n]$$

i.e., the posterior mean

$$\text{E} [\theta | Y_1, \dots, Y_n] = \int \theta f(\theta | Y_1, \dots, Y_n) d\theta.$$

Classical Bayesian estimators

Maximum a posteriori estimator

One chooses the 0/1 cost, defined for a small δ as

$$c(\theta, \hat{\theta}) = \begin{cases} 1, & \text{if } |\theta - \hat{\theta}| \geq \delta; \\ 0, & \text{otherwise.} \end{cases}$$

One shows that the estimator minimizing the corresponding risk is

$$\hat{\theta}_{\text{MAP}} = \underset{\theta}{\operatorname{argmax}} f(\theta | Y_1, \dots, Y_n)$$

i.e., the mode of the posterior distribution.

Posterior distribution

Classical Bayesian estimators

$$\hat{\theta}_{\text{MMSE}} = \int \theta f(\theta | Y_1, \dots, Y_n) d\theta$$

$$\hat{\theta}_{\text{MAP}} = \underset{\theta}{\operatorname{argmax}} f(\theta | Y_1, \dots, Y_n)$$

Thanks to the Bayes theorem, the posterior distribution writes

$$f(\theta | Y_1, \dots, Y_n) = \frac{f(Y_1, \dots, Y_n | \theta) f(\theta)}{f(Y_1, \dots, Y_n)}$$

and provides a comprehensive synthesis of the information provided by

- the **observations**, through the likelihood function $f(Y_1, \dots, Y_n | \theta)$
- the **knowledge available before any experience**, through the prior distribution $f(\theta)$

The quantity

$$f(Y_1, \dots, Y_n) = \int f(Y_1, \dots, Y_n | \theta) f(\theta) d\theta$$

is the **marginal likelihood** (or **evidence**), generally difficult to compute/evaluate.

Posterior distribution

Classical Bayesian estimators

$$\hat{\theta}_{\text{MMSE}} = \int \theta f(\theta | Y_1, \dots, Y_n) d\theta$$

$$\hat{\theta}_{\text{MAP}} = \underset{\theta}{\operatorname{argmax}} f(\theta | Y_1, \dots, Y_n)$$

Thanks to the Bayes theorem, the posterior distribution writes

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A short detour... via model comparison

On the use of the evidence

Problem statement

Let define the hypothesis problem defined by two concurrent probabilistic models

$$Y_i \sim \mathcal{M}_1 \quad \text{vs.} \quad Y_i \sim \mathcal{M}_2$$

One wants to decide between $\mathcal{M}_1$ and $\mathcal{M}_2$ (whatever their respective parameters).

Bayesian model comparison

Decision taken on the basis of the Bayes factor defined by

$$\kappa = \frac{f(Y_1, \dots, Y_n | \mathcal{M}_1)}{f(Y_1, \dots, Y_n | \mathcal{M}_2)} = \frac{P[\mathcal{M}_1 | Y_1, \dots, Y_n] P[\mathcal{M}_2]}{P[\mathcal{M}_2 | Y_1, \dots, Y_n] P[\mathcal{M}_1]}$$

where

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Which prior knowledge?

Or how to choose the prior distribution...

- **model-based** (i.e., handcraft) priors
 - ↪ to exploit a **prior knowledge** about the parameter to be estimated
 - sparsity, smoothness, support constraints (e.g., positivity)...
- **non-informative** (i.e., vague or objective) priors
 - ↪ a **paradox...** to enter into the *Bayesian framework*!
 - distribution defined by hyperparameters chosen to ensure a large variance
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A toy example (cont'd)

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One aims at estimating θ .

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$$\hat{\theta}_{\text{ML}} = \frac{1}{n} \sum_{i=1}^n Y_i$$

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Asymptotic behaviors

- When
 - $n \rightarrow 0$, no data
 - $\sigma^2 \rightarrow \infty$, no confidence in the data (high level of noise)
 - $\sigma_0^2 \rightarrow 0$, high level of confidence in the prior knowledge

then one trusts the prior knowledge

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Interpretation

- updating the prior knowledge $f(\theta)$ thanks to the data $Y_1, \dots, Y_n$

Benefits

- trade-off between prior knowledge and knowledge brought by the data
- possible variety of Bayesian estimators
- comprehensive description of the parameter space
- uncertainty quantification, here governed by the posterior variance

$$\eta^2 = \frac{\sigma^2 \sigma_0^2}{\sigma^2 + n \sigma_0^2} = \left(\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2} \right)^{-1}$$

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Credible intervals

Definition

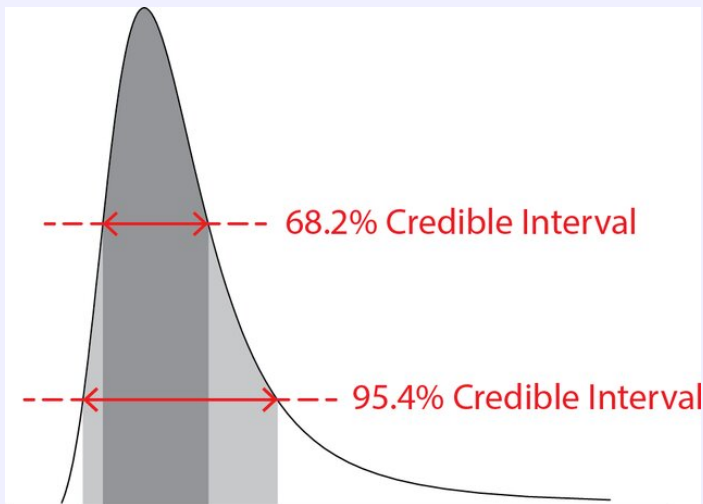
A posterior credible region with $100(1 - \alpha)\%$ confidence is a set $\mathbb{S}_\alpha$ such that

$$P[\theta \in \mathbb{S}_\alpha | Y_1, \dots, Y_n] = 1 - \alpha$$

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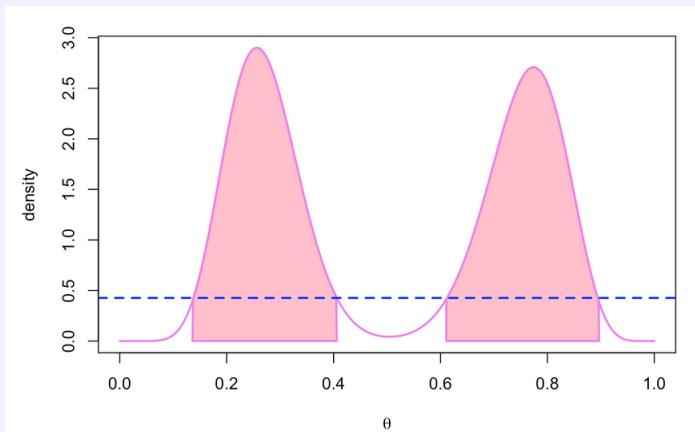
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Hierarchical Bayesian models

... or how to choose the hyperparameters?

Problem statement

The prior distribution $f(\theta|\phi)$ may depend on unknown hyperparameters ϕ .

Two strategies

- they are arbitrarily fixed to obtain an informative or, on the contrary, a noninformative prior
- one tries to estimate them

Three main approaches

- empirical Bayesian approach
→ estimating ϕ from training data
- frequentist approach
→ estimating ϕ following a maximum likelihood principle (e.g., algo. EM)
- “fully bayesian” approach
→ introducing a second level in the Bayesian hierarchy

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Principle

The hyperparameter ϕ is also considered as (the realisation of) a random variable, equipped with a prior distribution (hyperprior) $f(\phi)$.

Hierarchical Bayesian inference

Bayes theorem allows one to derive the joint posterior distribution of the unknown parameters and hyperparameters

$$f(\theta, \phi | Y_1, \dots, Y_n) \propto f(Y_1, \dots, Y_n | \theta) f(\theta | \phi) f(\phi)$$

where the normalizing constant is the inverse evidence $f(Y_1, \dots, Y_n)^{-1}$.

Joint estimation of θ et ϕ

$$\begin{aligned} \widehat{(\theta, \phi)}_{\text{MMSE}} &= \int (\theta, \phi)^T f(\theta, \phi | Y_1, \dots, Y_n) d\theta d\phi \\ \widehat{(\theta, \phi)}_{\text{MAP}} &= \underset{\theta, \phi}{\operatorname{argmax}} f(\theta, \phi | Y_1, \dots, Y_n) \end{aligned}$$

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Hierarchical Bayesian models

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STOP !

Outline

Estimation: generalities and “frequentist” approach

- Point estimation

- Maximum likelihood estimation

Bayesian estimation

- Toy example

- Bayesian paradigm

- Derivation of Bayesian estimators

- Key quantities

- Toy example (cont'd)

- Uncertainty quantification

- Hierarchical Bayesian models

Application to Bayesian inversion/regression

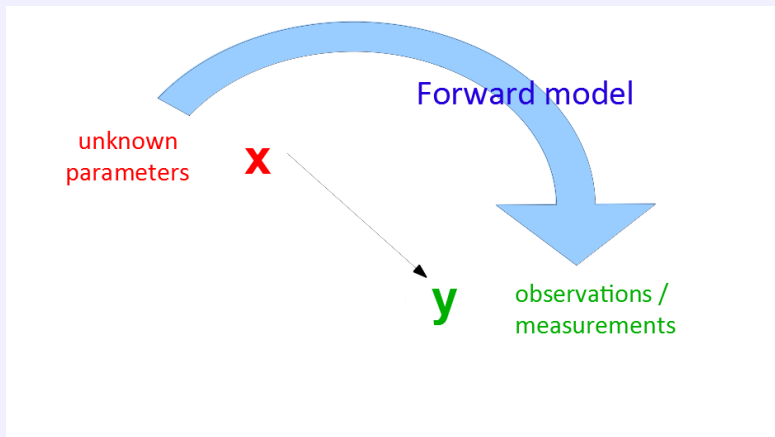
- Probabilistic formulation

- Bayesian regularization

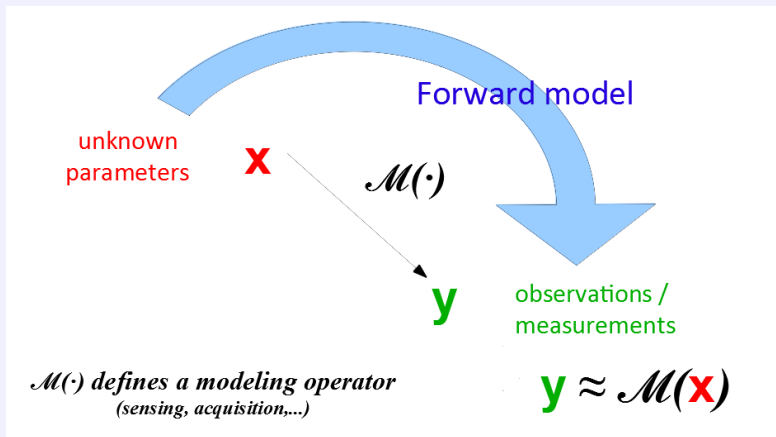
- Generalized Tikhonov regularization

Conclusion

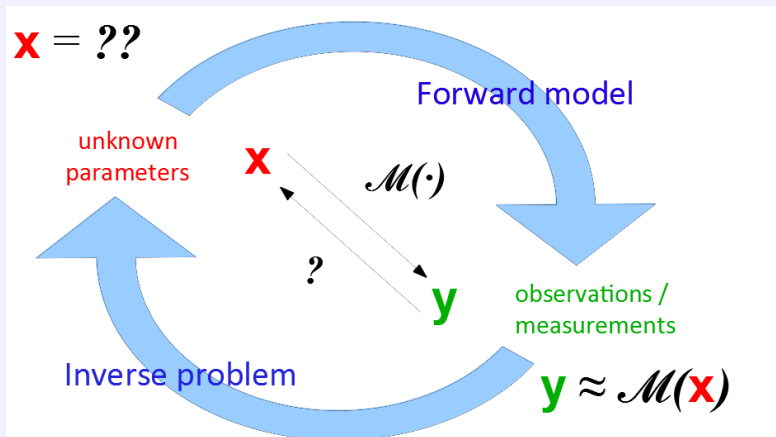
Inverse problems



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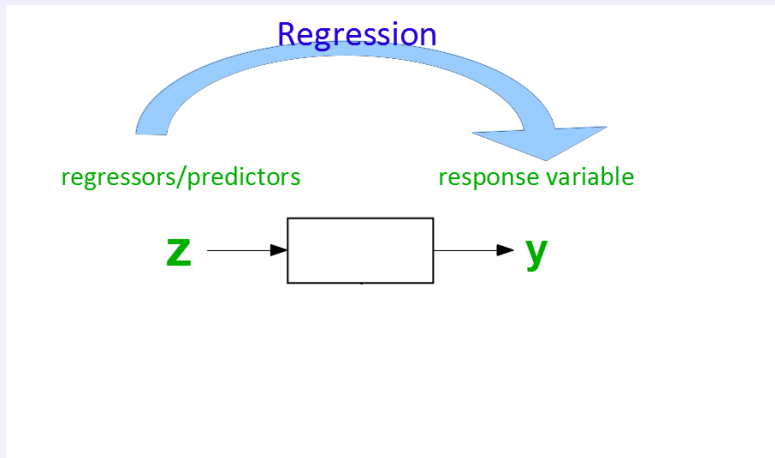
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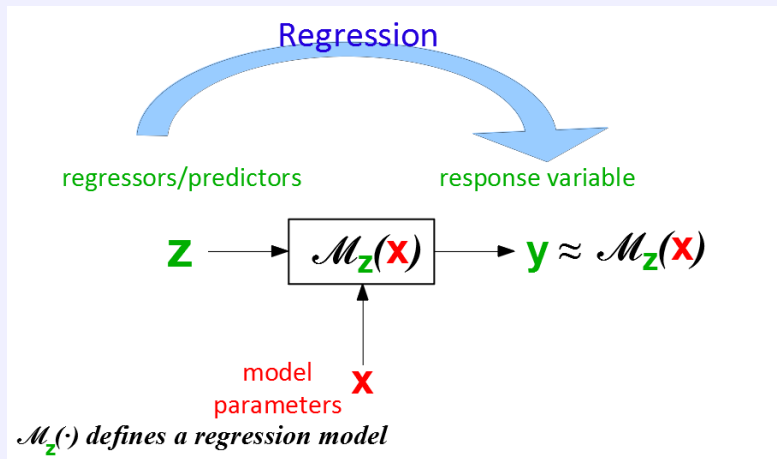
Regression



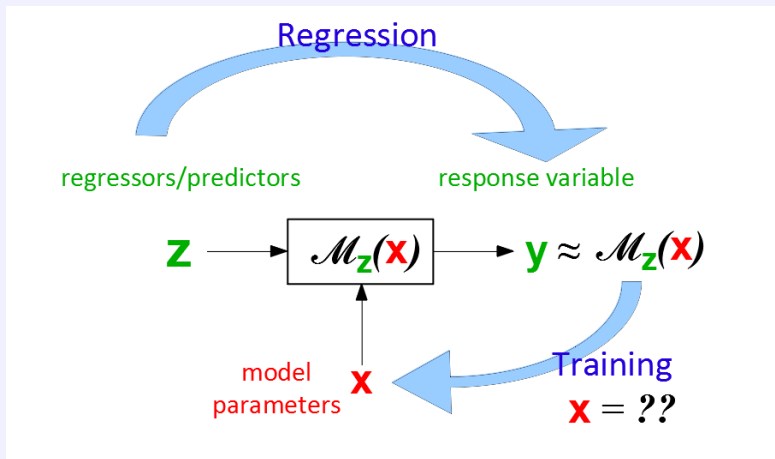
Regression



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Regression



Inverse, myopic and blind problems

$$\begin{aligned}\mathcal{H} : \quad \mathbb{R}^M &\rightarrow \mathbb{R}^N \\ \mathbf{x} &\mapsto \mathbf{y} \approx \mathcal{H}(\mathbf{x})\end{aligned}$$

Assumptions related to the forward operator $\mathcal{H}(\cdot)$

- fully known: **inverse** problem
 $\mathcal{H}(\cdot)$ non-invertible: ill-posed problem
 $\Rightarrow$ need for regularization
- partially known: **myopic** (semi-blind) problem
 $\mathcal{H}(\cdot)$ known through a parametrization $\mathcal{H}_\theta(\cdot)$, $\theta = [\theta_1, \dots, \theta_K]^T$
 $\Rightarrow$ joint estimation of $\mathbf{x}$ and θ
- fully unknown: **blind** problem
 $\Rightarrow$ need for additional assumptions supplémentaires
example: $\mathcal{H}(\cdot)$ is linear
 - deconvolution $y(n) \approx h(n) * x(n)$
 - blind source separation $\mathbf{y}(n) \approx \mathbf{H}\mathbf{x}(n)$
 - matrix factorization $\mathbf{Y} \approx \mathbf{H}\mathbf{X}$

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 $\Rightarrow$ joint estimation of $\mathbf{x}$ and θ
- fully unknown: **blind** problem
 $\Rightarrow$ need for additional assumptions supplémentaires
example: $\mathcal{H}(\cdot)$ is linear
 - deconvolution $y(n) \approx h(n) * x(n)$
 - blind source separation $\mathbf{y}(n) \approx \mathbf{H}\mathbf{x}(n)$
 - matrix factorization $\mathbf{Y} \approx \mathbf{H}\mathbf{X}$

Inverse, myopic and blind problems

$$\begin{aligned}\mathcal{H} : \quad \mathbb{R}^M &\rightarrow \mathbb{R}^N \\ \mathbf{x} &\mapsto \mathbf{y} \approx \mathcal{H}(\mathbf{x})\end{aligned}$$

Assumptions related to the forward operator $\mathcal{H}(\cdot)$

- fully known: **inverse** problem
 $\mathcal{H}(\cdot)$ non-invertible: ill-posed problem
 $\Rightarrow$ need for regularization
- partially known: **myopic** (semi-blind) problem
 $\mathcal{H}(\cdot)$ known through a parametrization $\mathcal{H}_{\theta}(\cdot)$, $\theta = [\theta_1, \dots, \theta_K]^T$
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Inversion and regularization

Bayesian formulation

How to choose a solution in the space of admissible solutions?

Inversion and regularization

Bayesian formulation

How to choose a solution in the space of admissible solutions?



introducing **penalties** and/or **constraints**
(driven by the target applicaiton)

Derivation of the criterion

- $\mathbf{y}|\mathbf{x} \sim f(\mathbf{y}|\mathbf{x})$: data-fitting term (driven by $\mathcal{H}$)
- $\mathbf{x} \sim f(\mathbf{x})$: **penalties** and **constraints**

Derivation of a Bayesian estimator from the *posterior* distribution

$$f(\mathbf{x}|\mathbf{y}) = \frac{1}{f(\mathbf{y})} f(\mathbf{y}|\mathbf{x}) f(\mathbf{x})$$

- $\hat{\mathbf{x}}_{\text{MMSE}} = \mathbb{E}[\mathbf{x}|\mathbf{y}] = \int \mathbf{x} f(\mathbf{x}|\mathbf{y}) d\mathbf{x}$
- $\hat{\mathbf{x}}_{\text{MAP}} = \operatorname{argmax}_{\mathbf{x}} f(\mathbf{x}|\mathbf{y})$

Linear forward model

One considers the observation model defined by

$$\mathbb{E}[\mathbf{y}|\mathbf{x}] = \mathbf{H}\mathbf{x}$$

where

- $\mathbf{x} \in \mathbb{R}^M$ is an unknown vector (to be estimated)
- $\mathbf{y} \in \mathbb{R}^N$ is the vector of the measurements/observations
- $\mathbf{H} \in \mathbb{R}^{N \times M}$ is a forward operator² (convolution, subsampling, etc...)

Probabilistic description of the noise/model errors

The target application yields a natural probabilistic model to build on $\mathbb{E}[\cdot]$

- additive (white) Gaussian noise

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{b} \quad \text{avec} \quad \mathbf{b} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\mathbf{b}})$$

- Poissonian noise (e.g., low-count imaging)

$$y_i | \mathbf{x} \sim \mathcal{P}([\mathbf{H}\mathbf{x}]_i), \quad i = 1, \dots, N$$

- multiplicative noise (e.g., speckle)

$$y_i = [\mathbf{H}\mathbf{x}]_i b_i, \quad i = 1, \dots, N \quad \text{avec} \quad b_i \sim \mathcal{G}(\alpha, 1/\alpha)$$

²Also defines a generalized linear model in a regression task.

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Gaussian linear inverse problem

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{b} \quad \text{with} \quad \mathbf{b} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\mathbf{b}})$$

The quantity $\mathbf{b}$ stands for the measurement noise and the modeling error

- with zero mean, $\mathbb{E}[\mathbf{b}] = \mathbf{0}$
- with covariance matrix $\Sigma_{\mathbf{b}}$

Problem statement

Recovering (i.e., estimating) $\mathbf{x}$ given the observations $\mathbf{y}$.

Likelihood function

Given the noise statistical model

$$f(\mathbf{y}|\mathbf{x}) = \left(\frac{1}{2\pi}\right)^{n/2} |\Sigma_{\mathbf{b}}|^{-1/2} \exp \left[-\frac{1}{2} (\mathbf{y} - \mathbf{H}\mathbf{x})^T \Sigma_{\mathbf{b}}^{-1} (\mathbf{y} - \mathbf{H}\mathbf{x}) \right]$$

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Gaussian linear inverse problem

Maximum likelihood estimation

Maximizing the (log-)likelihood leads to

$$\hat{\mathbf{x}}_{\text{ML}} = \underset{\mathbf{x}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_{\Sigma_{\mathbf{b}}^{-1}}^2 = \left(\mathbf{H}^T \Sigma_{\mathbf{b}}^{-1} \mathbf{H}\right)^{-1} \mathbf{H}^T \Sigma_{\mathbf{b}}^{-1} \mathbf{y}$$

Some particular cases

- $\Sigma_{\mathbf{b}} = \sigma_{\mathbf{b}}^2 \mathbf{I}$, then $(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T = \mathbf{H}^\dagger$ ($= \mathbf{H}^{-1}$ if $\mathbf{H}$ invertible)

$$\hat{\mathbf{x}}_{\text{ML}} = \underset{\mathbf{x}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_2^2 = \mathbf{H}^\dagger \mathbf{y} = \hat{\mathbf{x}}_{\text{LS}}$$

→ least-square solution

- $\Sigma_{\mathbf{b}} = \operatorname{diag}(\sigma_{\mathbf{b},1}^2, \dots, \sigma_{\mathbf{b},N}^2)$ then

$$\hat{\mathbf{x}}_{\text{ML}} = \underset{\mathbf{x}}{\operatorname{argmin}} \sum_{i=1}^N \frac{1}{\sigma_i^2} (y_i - [\mathbf{H}\mathbf{x}]_i)^2 = \hat{\mathbf{x}}_{\text{WLS}}$$

→ weighted least-square solution

Gaussian linear inverse problem

Maximum likelihood estimation

Issues with the ML/LS solution $\hat{\mathbf{x}}_{\text{ML}}$

- ill-posed problem, $(\mathbf{H}^T \Sigma_{\mathbf{b}}^{-1} \mathbf{H})$ non-invertible
- ill-conditioned problem (bad conditioning of $\mathbf{H}$): high sensibility w.r.t. to noise

Gaussian linear inverse problem

Maximum likelihood estimation

Example: denoising of smooth 1D signal

We are looking for the noise-free signal $\mathbf{x}$

$$\mathbf{y} = \mathbf{x} + \mathbf{n}$$

- $M = N$
- the operator $\mathbf{H}$ is the identity $\mathbf{I}$
- $\mathbf{H}^\dagger = \mathbf{H}^{-1} = \mathbf{I}$

Least-square solution

$$\hat{\mathbf{x}}_{\text{LS}} = \operatorname{argmin}_{\mathbf{x}} \|\mathbf{y} - \mathbf{x}\|_2^2$$

$\Downarrow$

$$\hat{\mathbf{x}}_{\text{LS}} = \mathbf{y}$$

(Bayesian) regularization is required!

Gaussian linear inverse problem

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Prior model

The unknown vector $\mathbf{x}$ is given a prior distribution $f(\mathbf{x}|\phi)$, herein chosen as a (conjugate) Gaussian distribution $\mathbf{x}|\phi \sim \mathcal{N}(\mu_{\mathbf{x}}, \Sigma_{\mathbf{x}})$:

$$f(\mathbf{x}|\phi) = \left(\frac{1}{2\pi}\right)^{m/2} |\Sigma_{\mathbf{x}}|^{-1/2} \exp \left[-\frac{1}{2} (\mathbf{x} - \mu_{\mathbf{x}})^T \Sigma_{\mathbf{x}}^{-1} (\mathbf{x} - \mu_{\mathbf{x}}) \right]$$

with $\phi = \{\mu_{\mathbf{x}}, \Sigma_{\mathbf{x}}\}$ the set of hyperparameters.

Gaussian linear inverse problem Bayesian estimation

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Gaussian linear inverse problem

Bayesian estimation

Posterior distribution

Thanks to Bayes' rule $f(\mathbf{x}|\mathbf{y}, \phi) \propto f(\mathbf{y}|\mathbf{x}) f(\mathbf{x}|\phi)$, one shows that

$$\mathbf{x}|\mathbf{y}, \phi \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{x}|\mathbf{y}}, \boldsymbol{\Sigma}_{\mathbf{x}|\mathbf{y}})$$

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ML, MMSE, MAP estimators

Let consider the particular case

$$\boldsymbol{\Sigma}_{\mathbf{b}} = \sigma_{\mathbf{b}}^2 \mathbf{I} \quad (\text{white noise})$$

$$\boldsymbol{\Sigma}_{\mathbf{x}} = \sigma_{\mathbf{x}}^2 \mathbf{I}$$

then

$$\hat{\mathbf{x}}_{\text{ML}} = \left(\mathbf{H}^T \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{y}$$

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with $\lambda = \frac{\sigma_{\mathbf{b}}^2}{\sigma_{\mathbf{x}}^2} (> 0)$.

Remark

Invertibility of $(\mathbf{H}^T \mathbf{H} + \lambda \mathbf{I})$ ensures once λ is sufficiently high...

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Gaussian linear inverse problem

MAP estimation

Maximizing the (log-)posterior distribution yields

$$\hat{\mathbf{x}}_{\text{MAP}} = \underset{\mathbf{x}}{\operatorname{argmin}} \underbrace{\log f(\mathbf{y}|\mathbf{x}, \phi)}_{\text{data-fitting}} + \underbrace{\log f(\mathbf{x}|\phi)}_{\text{regularization}}$$

Here

$$\hat{\mathbf{x}}_{\text{MAP}} = \underset{\mathbf{x}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_2^2 + \lambda \|\mathbf{x} - \boldsymbol{\mu}_{\mathbf{x}}\|_2^2$$

→ minimization a ℓ_2 - ℓ_2 criterion (Tikhonov regularization, aka “*ridge regression*”)

Advantage

The Bayesian “framework” provides a probabilistic interpretation of the regularization parameter

$$\lambda = \frac{\sigma_{\mathbf{b}}^2}{\sigma_{\mathbf{x}}^2}.$$

Remark

$\hat{\theta}_{\text{MAP}} = \hat{\theta}_{\text{MMSE}}$ for symmetric posterior distributions...

Gaussian linear inverse problem

MAP estimation

$$\hat{\mathbf{x}}_\lambda \in \underset{\mathbf{x}}{\operatorname{argmin}} \mathcal{J}_\lambda(\mathbf{x})$$

with $\mathcal{J}_\lambda(\mathbf{x}) = \mathcal{D}(\mathbf{y}|\mathbf{H}\mathbf{x}) + \lambda\mathcal{P}(\mathbf{x})$

- when $\lambda \rightarrow 0$ then

$$\hat{\mathbf{x}}_\lambda \rightarrow \underset{\mathbf{x}}{\operatorname{argmin}} \mathcal{D}(\mathbf{y}|\mathbf{H}\mathbf{x})$$

→ no prior knowledge, no regularization!

- when $\lambda \rightarrow \infty$ then

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Gaussian linear inverse problem

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Tikhonov regularization: generalized form

Historical and ubiquitous regularization

- initially stated in the functional case (infinite dimension) by Tikhonov (1943) and Phillips (1962)
- subsequently extended to the finite dimensional case (from a statistical point-of-view)

Definition

The regularization (or penalization) is defined as

$$\mathcal{P}(\mathbf{x}) = \|\Gamma(\mathbf{x} - \boldsymbol{\mu})\|_2^2$$

where $\Gamma \in \mathbb{R}^{\tilde{M} \times M}$ (e.g., $\tilde{M} \leq M$) and $\boldsymbol{\mu} \in \mathbb{R}^M$ are fixed parameters.

Bayesian formulation

The prior distribution is chosen as a multivariate Gaussian distribution

$$p(\mathbf{x}) \propto \exp \left[-\frac{1}{2} \|\Gamma(\mathbf{x} - \boldsymbol{\mu})\|_2^2 \right]$$

Geometric interpretation

Mahalanobis distance

When $\Lambda^{-1} = \Gamma^T \Gamma \in \mathbb{R}^{M \times M}$ is a positive (semi-)definitive matrix, then the regularization $\mathcal{P}(\mathbf{x})$ can be rewritten as

$$\begin{aligned}\mathcal{P}(\mathbf{x}) &= \|\Gamma(\mathbf{x} - \boldsymbol{\mu})\|_2^2 \\ &= \langle \mathbf{x} - \boldsymbol{\mu}, \Lambda^{-1}(\mathbf{x} - \boldsymbol{\mu}) \rangle \\ &= \langle \mathbf{x} - \boldsymbol{\mu}, \mathbf{x} - \boldsymbol{\mu} \rangle_{\Lambda^{-1}} \\ &= d^2(\mathbf{x}, \boldsymbol{\mu})\end{aligned}$$

where $d(\mathbf{x}, \mathbf{m})$ is the **Mahalanobis distance** between $\mathbf{x}$ and $\boldsymbol{\mu}$ induced by the inner-product $\langle \cdot, \cdot \rangle_{\Lambda^{-1}}$

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\Lambda^{-1}} = \mathbf{u}^T \Lambda^{-1} \mathbf{v}$$

Choosing the Tikhonov regularization promotes solutions $\mathbf{x}$ close to $\boldsymbol{\mu}$
(in the Mahalanobis distance sense)

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Special cases (of interest)

- diagonal: if $\Gamma = \text{diag}(\gamma_1, \dots, \gamma_M)$ then

$$\mathcal{P}(\mathbf{x}) = \sum_{m=1}^M \gamma_m^2 (x_m - \mu_m)^2$$

→ weighted distance

→ may account for an **anisotropic** space (with different dynamics)

- identity: if $\Gamma = \mathbb{I}_M$ then

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- if $\mu = \mathbf{0}$ and $\Gamma = \mathbf{I}_M$, then the regularization is

$$\mathcal{P}(\mathbf{x}) = \|\mathbf{m}\|_2^2$$

→ promotes **solutions of low (ℓ_2 -)energy**

- if $\mu = \mathbf{0}$ and Γ is a smoothing operator, e.g.,

$$\Gamma = \begin{pmatrix} -1 & 2 & -1 & 0 & \dots & 0 \\ 0 & -1 & 2 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & -1 & 2 & -1 \end{pmatrix}, \text{ i.e., } \Gamma \mathbf{x} = \begin{pmatrix} -x_1 + 2x_2 - x_3 \\ -x_2 + 2x_3 - x_4 \\ \vdots \\ -x_{M-2} + 2x_{M-1} - x_M \end{pmatrix}$$

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$\ell_2 - \ell_2$ linear inverse problem

Quadratic data fitting term + Thikonov regularization

Main ingredients

Forward model	$\mathbf{y} \approx \mathbf{H}\mathbf{x}$	(linear model)
Data-fitting term	$\mathcal{D}(\cdot \cdot) = \ \cdot - \cdot\ _2^2$	(quadratic cost)
Thikonov regularization	$\mathcal{P}(\mathbf{x}) = \ \mathbf{\Gamma}(\mathbf{x} - \boldsymbol{\mu})\ _2^2$	(quadratic regularization)

Problem formulation

We want to solve $\hat{\mathbf{x}}_\lambda = \operatorname{argmin}_{\mathbf{x}} \mathcal{J}_\lambda(\mathbf{x})$
with $\mathcal{J}_\lambda(\mathbf{x}) = \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_2^2 + \lambda \|\mathbf{\Gamma}(\mathbf{x} - \boldsymbol{\mu})\|_2^2$

Resolution

It amounts to solve $\nabla \mathcal{J}_\lambda(\mathbf{x})|_{\mathbf{x}=\hat{\mathbf{x}}_\lambda} = 0$

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Data-fitting term	$\mathcal{D}(\cdot \cdot) = \ \cdot - \cdot\ _2^2$	(quadratic cost)
Thikonov regularization	$\mathcal{P}(\mathbf{x}) = \ \mathbf{\Gamma}(\mathbf{x} - \boldsymbol{\mu})\ _2^2$	(quadratic regularization)

Problem formulation

We want to solve $\hat{\mathbf{x}}_\lambda = \operatorname{argmin}_{\mathbf{x}} \mathcal{J}_\lambda(\mathbf{x})$
with $\mathcal{J}_\lambda(\mathbf{x}) = \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_2^2 + \lambda \|\mathbf{\Gamma}(\mathbf{x} - \boldsymbol{\mu})\|_2^2$

Resolution

It amounts to solve $\nabla \mathcal{J}_\lambda(\mathbf{x})|_{\mathbf{x}=\hat{\mathbf{x}}_\lambda} = 0$

$\ell_2 - \ell_2$ linear inverse problem

Quadratic data fitting term + Thikonov regularization

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Limit cases

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Special case: $\mu = 0$

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Least-square solution

$$\hat{\mathbf{x}}_{\text{LS}} = \left[\mathbf{H}^T \mathbf{H} \right]^{-1} \mathbf{H}^T \mathbf{y}$$

ℓ_2 -regularized solution

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Improved conditioning

- the LS-solution relies on the eigenvalue values of $\mathbf{H}^T \mathbf{H}$
- the ℓ_2 -regularized solution relies on the eigenvalues of $\mathbf{H}^T \mathbf{H} + \lambda \mathbf{\Gamma}^T \mathbf{\Gamma}$

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Spectral analysis

$$\text{eigenvalue}(\mathbf{H}^T \mathbf{H} + \lambda \mathbf{I}) = \text{eigenvalue}(\mathbf{H}^T \mathbf{H}) + \lambda$$

- for $\lambda > \min \{ \text{eigenvalue}(\mathbf{H}^T \mathbf{H}) \}$, $\min \{ \text{eigenvalue}(\mathbf{H}^T \mathbf{H} + \lambda \mathbf{I}) \} > 0$
ill-posed problem $\rightsquigarrow$ well-posed problem
- for $\epsilon > 0$, it exists $\lambda > 0$ such that $\min \{ \text{eigenvalue}(\mathbf{H}^T \mathbf{H} + \lambda \mathbf{I}) \} > \epsilon$
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Example: denoising of smooth 1D signal

Observation model

We are looking for the noisy signal $\mathbf{x}$, assumed to be smooth,

$$\mathbf{y} = \mathbf{x} + \mathbf{n}$$

- $M = N$
- the operator $\mathbf{H}$ is the identity $\mathbf{I}$
- $\mathbf{H}^\dagger = \mathbf{H}^{-1} = \mathbf{I}$

Least-square solution

$$\hat{\mathbf{x}}_{\text{LS}} = \operatorname{argmin}_{\mathbf{x}} \|\mathbf{y} - \mathbf{x}\|_2^2$$

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$$\hat{\mathbf{x}}_{\text{LS}} = \mathbf{y}$$

Regularization is required!

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Example: denoising of smooth 1D signal

Smoothing regularization

The matrix Γ is chosen as a smoothing operator,

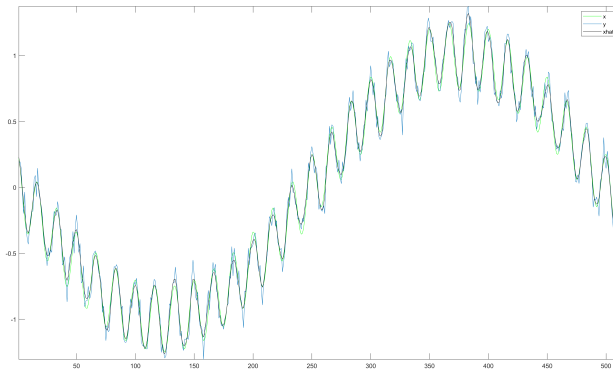
$$\Gamma = \begin{pmatrix} -1 & 2 & -1 & 0 & \dots & 0 \\ 0 & -1 & 2 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & -1 & 2 & -1 \end{pmatrix} \in \mathbb{R}^{(M-2) \times M}$$

which leads to

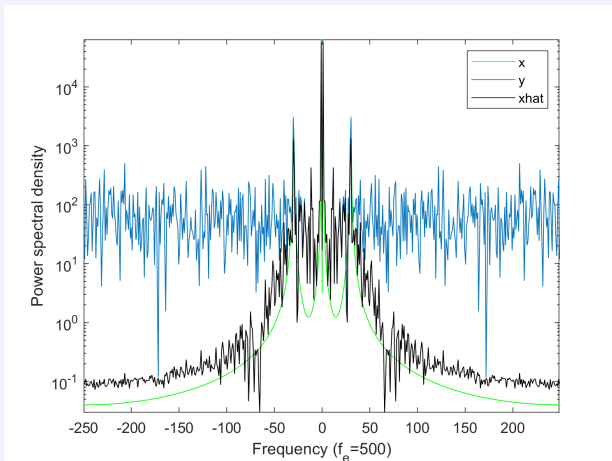
$$\begin{aligned} \hat{\mathbf{x}}_{\lambda} &= \operatorname{argmin}_{\mathbf{x}} \|\mathbf{y} - \mathbf{x}\|_2^2 + \|\Gamma \mathbf{x}\|_2^2 \\ &\Downarrow \\ \hat{\mathbf{x}}_{\lambda} &= [\mathbf{I} + \lambda \Gamma^T \Gamma]^{-1} \mathbf{y} \end{aligned}$$

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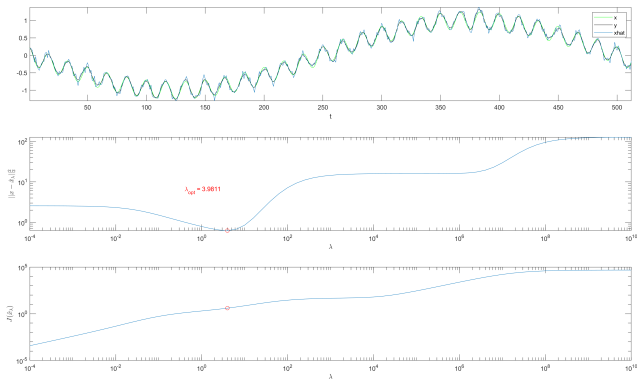
Results (for “optimal” λ)



Example: denoising of smooth 1D signal Results (for “optimal” λ)

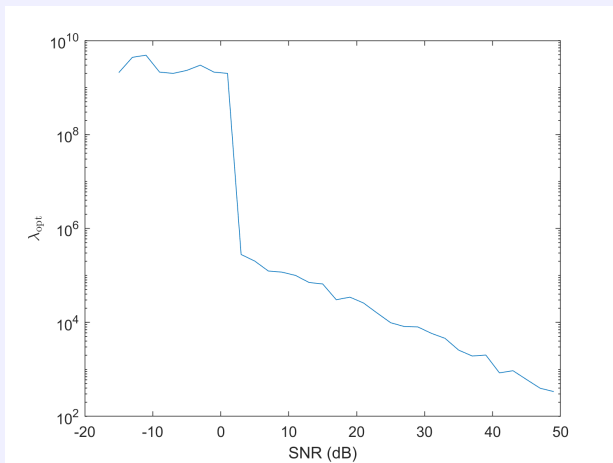


Example: denoising of smooth 1D signal As a function of λ



Example: denoising of smooth 1D signal

As a function of the SNR



Outline

Estimation: generalities and “frequentist” approach

- Point estimation

- Maximum likelihood estimation

Bayesian estimation

- Toy example

- Bayesian paradigm

- Derivation of Bayesian estimators

- Key quantities

- Toy example (cont'd)

- Uncertainty quantification

- Hierarchical Bayesian models

Application to Bayesian inversion/regression

- Probabilistic formulation

- Bayesian regularization

- Generalized Tikhonov regularization

Conclusion

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Bayesian inference

What?

A smart, elegant way

- to describe a prior knowledge within a probabilistic fashion
- to balance information brought by data and prior knowledge

How?

- by building a probabilistic model of the observations
- by considering unknown parameters as random

Why?

- to update the prior knowledge
- to regularize ill-posed and/or ill-conditioned problems
- to interpret (and estimate) regularization parameters (or hyperparameters)
- to derive relevant estimators wrt an application (beyond LS/ML estimator)
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Remaining challenges

Challenges

- Choosing the hyperparameters ϕ defining the *prior* model

Opportunities

- Introducing a second level in the Bayesian hierarchy

$$\phi \sim f(\phi)$$

then marginalizing

$$f(\mathbf{x}|\mathbf{y}) = \frac{1}{f(\mathbf{y})} \int f(\mathbf{y}|\mathbf{x}) f(\mathbf{x}|\phi) f(\phi) d\phi$$

Conclusion

Remaining challenges

Challenges

- Choosing the hyperparameters ϕ defining the *prior* model
- Optimizing (MAP) or integrating (MMSE) the resulting (log-)posterior distribution

Opportunities

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- Resorting to numerical algorithms
 - optimization algorithms to approximate $\hat{\mathbf{x}}_{\text{MAP}}$
 - Monte Carlo sampling to approximate $\hat{\mathbf{x}}_{\text{MMSE}}$ (and credible regions)

$$\hat{\mathbf{x}}_{\text{MMSE}} = \mathbb{E}[\mathbf{x}|\mathbf{y}] \approx \frac{1}{N_{\text{MC}}} \sum_{t=1}^{N_{\text{MC}}} \tilde{\mathbf{x}}^{(t)} \quad \text{avec} \quad \tilde{\mathbf{x}}^{(t)} \sim f(\mathbf{x}|\mathbf{y})$$

Bayesian methods for uncertainty quantification

Part I – Bayesian inference

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